

SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

THIS MONTH IN SCIENCE:

In Space Age, learn about the protocols in place when dealing with alien life. Then, find out how a bicycle works in Ignorance. After that learn about the techniques being used to translate ancient and unknown languages and more...



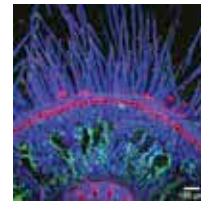
Eyes on Sentinel

The Sentinel-6 Michael Freilich, an instrument dedicated to investigating ocean surface topography, is the latest satellite to be added to NASA's Eyes on the Solar System web app, allowing you to follow the satellite over its mission, and view it in context of other spacecraft. <https://digit.in/sentinel6>

WHAT'S NEW

Microbiome census of the soil

This picture shows the roots of maize plants in blue, that are colonised by bacteria indicated with pink. The bacteria and fungi living in soil are called microbial dark matter, because it is very difficult to replicate and study them in laboratory environments. These mysterious organisms occupy the soil microbiome, influencing the overall health of the plant starting from the roots. The bacteria even offer protection to the plants from pathogenic fungi. At times, houseplants can be seen to wither and die even after careful tending, and it may be related to the changes in the soil microbiome. Researchers from Caltech have developed a new technology that can analyse



DNA collected from soil samples, and determine what kind of organisms live in the soil, and even what specific abilities they have. The approach can detect for example, if any

of the bacteria can produce antifungal molecules. The researchers hope that by understanding the interactions of the microbes and the plants within the soil environment, it may be possible to develop strategies to increase the yield of crops, make them more resilient to diseases, and ultimately create more sustainable sources of food. <https://digit.in/plntprb>

Largest Arctic expedition returns to the home port

The icebreaker Polarstern docked into port in Bremerhaven, Germany, marking an end to the Multidisciplinary drifting Observatory for the Study of Arctic Climate (MOSAiC) expedition, that had been ongoing for over a year. In 2019, the ship ventured into the central Arctic and became embedded in the ice, then began drifting across the north pole, completely controlled by natural forces. While most of the world came to



a standstill because of the pandemic, the work on MOSAiC continued without being affected. The expedition was supported by seven additional ships, and several aircraft. The research

ship ventured further north than any ship before, and gathered critical information from a region undergoing the most pronounced changes due to climate change. The ship has gathered more than 100 terabytes of data over the voyage, and scientists have already begun to analyse the data; it will take years to get through it all. It is unlikely that another expedition of the same scale will be mounted for years. <https://digit.in/mosaic>



CERN releases LHC schedule

CERN has just revealed the timelines for wrapping up the upgrades and repairs on the LHC, and when the experiments are slated to begin. There has been a delay of around four months because of the pandemic, and the accelerator is expected to start working in September 2021. <https://digit.in/lhc21>



Studying minerals from orbit

It is not for the purposes of mining, but instead to study the evolution of the solar system. Researchers have developed technologies that can probe the chemical composition of olivine, a mineral found on all solar system objects, using instruments deployed in orbit around the body. <https://digit.in/olivine>



Tractor beams for lightning

Scientists in Australia have come up with the rather strange idea of using beams of lasers as lightning rods, in an effort to prevent bushfires. The laser beam performs the same function as lightning, and creates a path for an electrical charge, directing it towards a suitable target. <https://digit.in/liitrac>